Experiment 09

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%--------------------------------------------------------------------
% Bode Plot
%--------------------------------------------------------------------
clc;
clear;
close all;
% Define the Laplace variable
s = tf('s');
% Plant transfer function
G = 1/(s*(s+1)*(0.5*s+1));
% Plot the uncompensated Bode plot
figure;
margin(G);
grid on;
title('Bode Plot of Uncompensated System');
% Find gain crossover frequency and phase margin
[Gm, Pm, Wcg, Wcp] = margin(G);
fprintf('Uncompensated Phase Margin = %f degrees\n', Pm);
fprintf('Gain Crossover Frequency = %f rad/s\n', Wcp);
% Desired phase margin
PM_desired = 45;
% Choose a lag compensator
beta = 10;
T = 1/(Wcp/10);
% Lag compensator transfer function
Gc = (T*s + 1)/(beta*T*s + 1);
% Compensated open-loop transfer function
G_comp = Gc*G;
% Plot compensated and uncompensated Bode plots
figure;
margin(G);
hold on;
margin(G_comp);
grid on;
legend('Uncompensated System','Lag Compensated System');
title('Bode Plot Before and After Lag Compensation');
% Display lag compensator
disp('Lag Compensator Gc(s) = ');
Gc
% Display compensated system
disp('Compensated Open-Loop Transfer Function = ');
G_comp
%--------------------------------------------------------------------
% Root Locus
%--------------------------------------------------------------------
clc;
clear;
close all;
% Define Laplace variable
s = tf('s');
% Uncompensated plant (K is handled by rlocus)
G = 1/(s*(s+1)*(s+3));
% Root locus of uncompensated system
figure;
rlocus(G);
grid on;
title('Root Locus of Uncompensated System');
```

```
% Lag compensator parameters
beta = 10;
T = 5;

% Lag compensator: (Ts + 1)/(beta*Ts + 1)
Gc = (T*s + 1)/(beta*T*s + 1);

% Compensated open-loop transfer function
G_comp = G*Gc;
% Root locus of compensated system
figure;
rlocus(G_comp);
grid on;
title('Root Locus with Lag Compensation');
% Display transfer functions
disp('Uncompensated Plant:');
G
disp('Lag Compensator:');
Gc
disp('Compensated Open-Loop Transfer Function:');
G_comp
```

Experminent 10

```
%--------------------------------------------------------------------
% Bode Plot
%--------------------------------------------------------------------

clc;
clear;
close all;

s = tf('s');

% Plant transfer function
G = 1/(s*(s+1)*(s+2));

% Bode plot of uncompensated system
figure;
margin(G);
grid on;
title('Bode Plot of Uncompensated System');

% Lag compensator parameters
beta = 10;
Tlag = 2;

Glag = (Tlag*s + 1)/(beta*Tlag*s + 1);

% Lead compensator parameters
alpha = 0.1;
Tlead = 1;

Glead = (Tlead*s + 1)/(alpha*Tlead*s + 1);
```

```
% Combined lag-lead compensator
Gc = Glag*Glead;

% Compensated system
Gcomp = G*Gc;

% Compare Bode plots
figure;
margin(G);
hold on;
margin(Gcomp);
grid on;
legend('Uncompensated','Lag-Lead Compensated');
title('Bode Plot with Lag-Lead Compensation');

disp('Lag Compensator:');
Glag

disp('Lead Compensator:');
Glead

disp('Combined Compensator:');
Gc

%--------------------------------------------------------------------
% Root Locus
%--------------------------------------------------------------------


clc;
clear;
close all;

s = tf('s');

% Plant transfer function
G = 4*s/(s + 0.5);

% Root locus plot
figure;
rlocus(G);
grid on;
title('Root Locus of G(s) = 4s/(s + 0.5)');

% Display transfer function
disp('Given Transfer Function:');
G
```